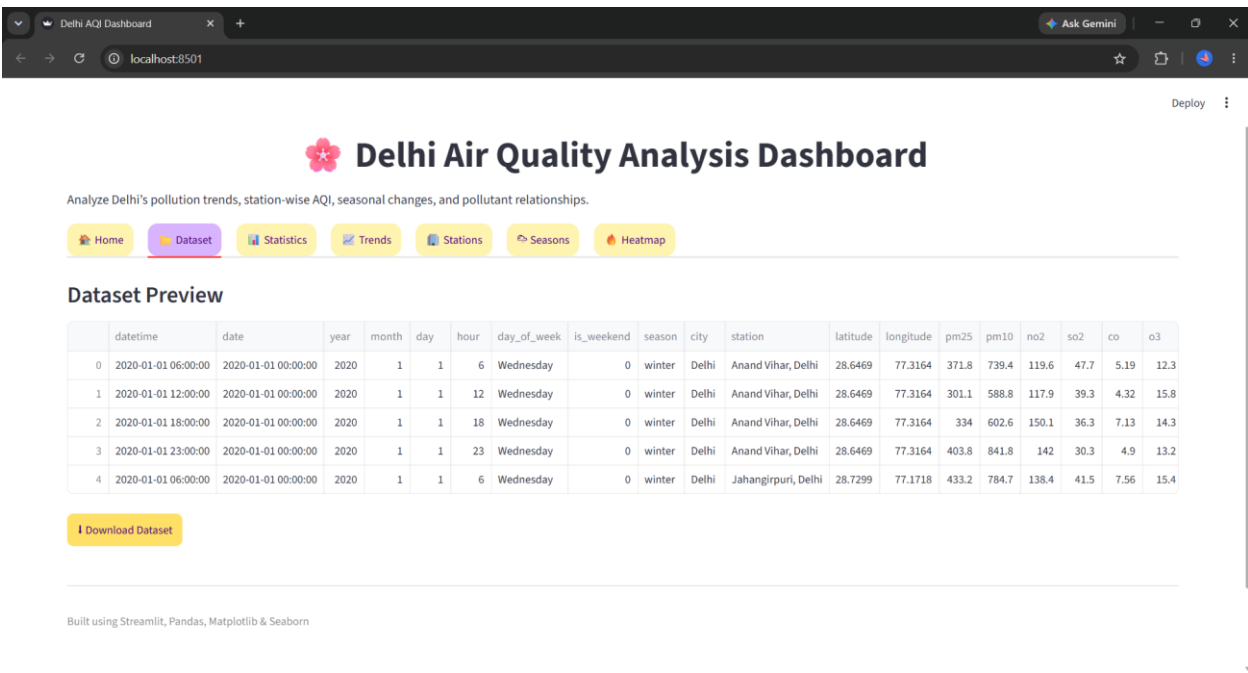


DELHI AIR QUALITY ANALYSIS DASHBOARD

Air Quality Index (AQI) is a numerical value used to measure the quality of air and its impact on human health. Higher AQI values indicate poor air quality and greater health risks. This project focuses on analyzing Delhi's air pollution using real-world AQI data. An interactive dashboard was developed using Streamlit to visualize AQI trends, station-wise pollution levels, seasonal changes, and pollutant relationships.

DATASET OVERVIEW

The dataset used for this project contains air quality data collected from multiple monitoring stations across Delhi. It includes pollutant measurements such as PM2.5, PM10, NO2, SO2, CO, and O3 along with AQI values, AQI categories, weather conditions, and seasonal information. The dataset contains more than 200,000 records and 25 columns, making it suitable for detailed analysis.

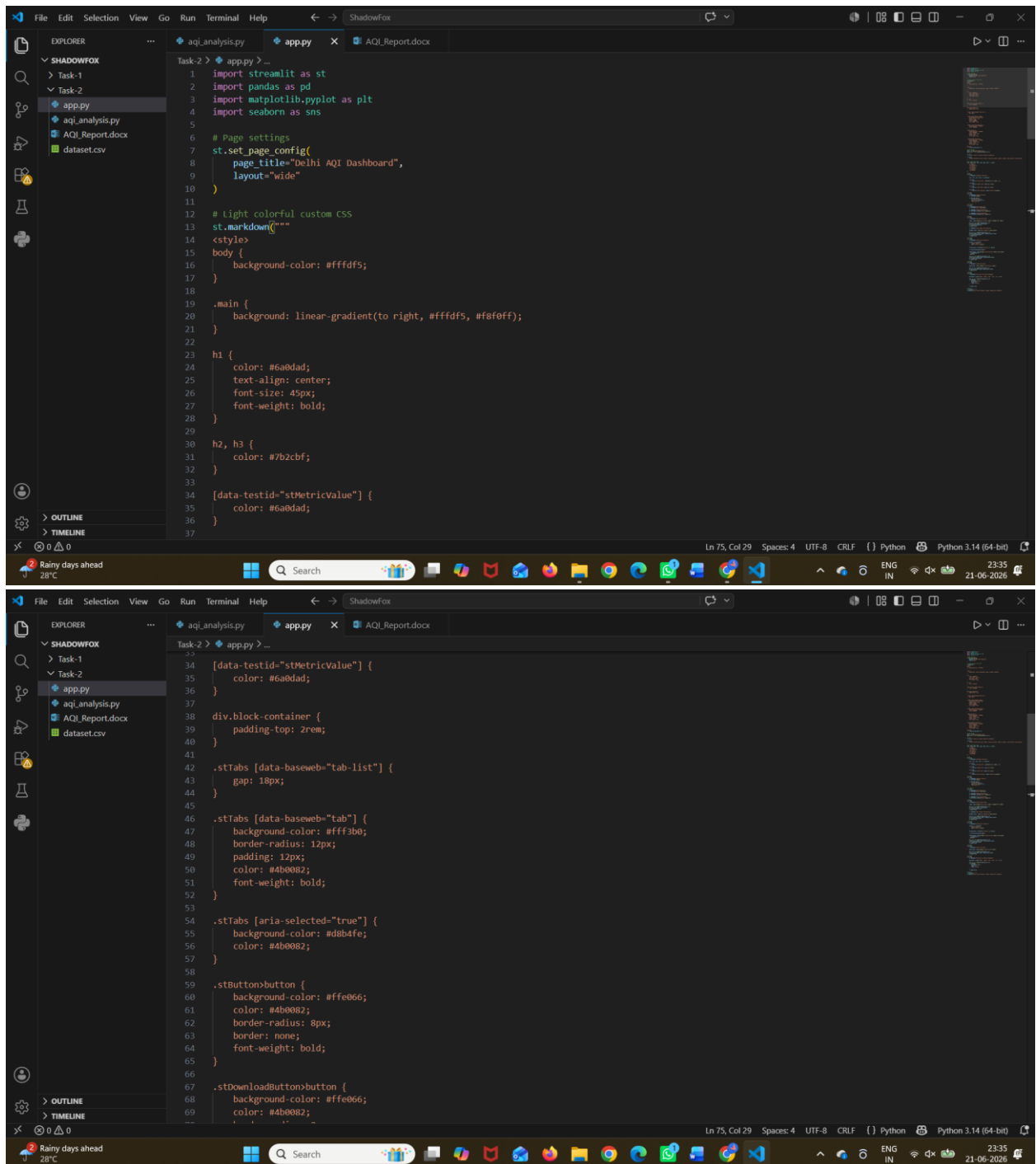


DASHBOARD DEVELOPMENT

The dashboard was developed using Streamlit, Pandas, Matplotlib, and Seaborn. Streamlit was used to create an interactive web application for analyzing Delhi AQI data. Pandas was used for data handling and preprocessing, while Matplotlib and Seaborn were used for visualizations.

CODE IMPLEMENTATION

Code:

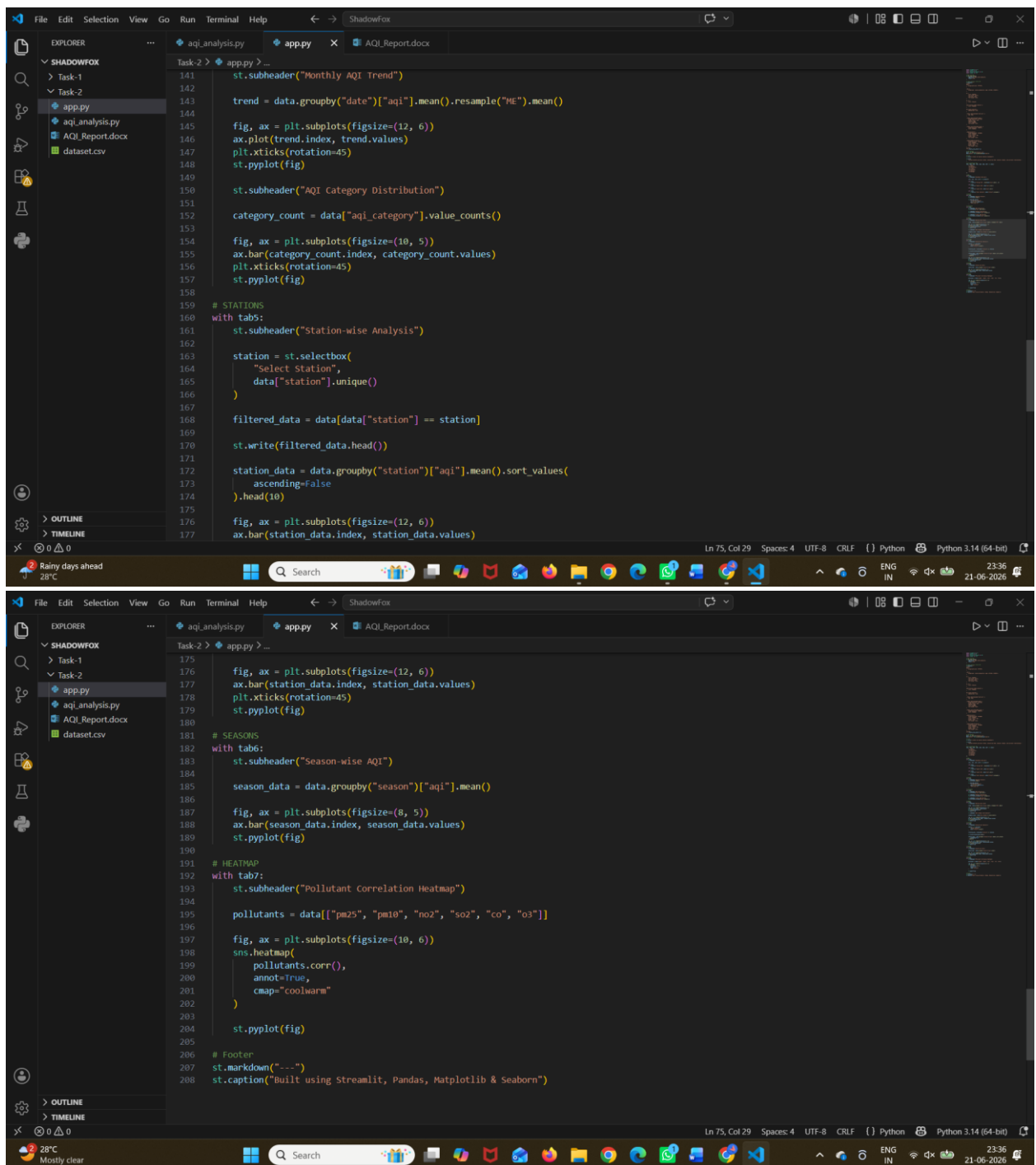


```
Task-2 > app.py > ...
1 import streamlit as st
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 # Page settings
7 st.set_page_config(
8     page_title="Delhi AQI Dashboard",
9     layout="wide"
10 )
11
12 # Light colorful custom CSS
13 st.markdown("""
14 <style>
15 body {
16     background-color: #ffffdf;
17 }
18
19 .main {
20     background: linear-gradient(to right, #ffffdf, #fafaff);
21 }
22
23 h1 {
24     color: #6a0dad;
25     text-align: center;
26     font-size: 45px;
27     font-weight: bold;
28 }
29
30 h2, h3 {
31     color: #7b2cbf;
32 }
33
34 [data-testid="stMetricValue"] {
35     color: #6a0dad;
36 }
37
38 [data-testid="stMetricValue"] {
39     color: #6a0dad;
40 }
41
42 div.block-container {
43     padding-top: 2rem;
44 }
45
46 .stTabs [data-baseweb="tab-list"] {
47     gap: 18px;
48 }
49
50 .stTabs [data-baseweb="tab"] {
51     background-color: #fff3b0;
52     border-radius: 12px;
53     padding: 12px;
54     color: #4b0082;
55     font-weight: bold;
56 }
57
58 .stTabs [aria-selected="true"] {
59     background-color: #d8b4fe;
60     color: #4b0082;
61 }
62
63 .stButton>button {
64     background-color: #ffe066;
65     color: #4b0082;
66     border-radius: 8px;
67     border: none;
68     font-weight: bold;
69 }
70
71 .stDownloadButton>button {
72     background-color: #ffe066;
73     color: #4b0082;
74 }
```

```
File Edit Selection View Go Run Terminal Help Shadowfox
EXPLORER
SHADOWFOX
Task-1
Task-2
app.py
aqi_analysis.py
AQI_Report.docx
dataset.csv

Task-2 > app.py > ...
69 color: #4b0082;
70 border-radius: 8px;
71 border: none;
72 font-weight: bold;
73 }
74 </style>
75 <div>
76 </div>
77 # Load dataset
78 data = pd.read_csv("dataset.csv")
79 data["date"] = pd.to_datetime(data["date"])
80
81 # Title
82 st.title("🌸 Delhi Air Quality Analysis Dashboard")
83
84 st.write(
85     "Analyze Delhi's pollution trends, station-wise AQI, seasonal changes, and pollutant relationships."
86 )
87
88 # Top navigation tabs
89 tab1, tab2, tab3, tab4, tab5, tab6, tab7 = st.tabs([
90     "Home",
91     "Dataset",
92     "Statistics",
93     "Trends",
94     "Stations",
95     "Seasons",
96     "Heatmap"
97 ])
98
99 # HOME
100 with tab1:
101     st.subheader("Dashboard Overview")
102
103     col1, col2, col3, col4 = st.columns(4)
104
105     with col1:
```

```
106         with col1:
107             st.metric("Average AQI", round(data["aqi"].mean(), 2))
108
109         with col2:
110             st.metric("Highest AQI", data["aqi"].max())
111
112         with col3:
113             st.metric("Lowest AQI", data["aqi"].min())
114
115         with col4:
116             st.metric("Total Stations", data["station"].nunique())
117
118 # DATASET
119 with tab2:
120     st.subheader("Dataset Preview")
121     st.write(data.head())
122
123     st.download_button(
124         "Download Dataset",
125         data.to_csv(index=False),
126         "delhi_aqi.csv"
127     )
128
129 # STATISTICS
130 with tab3:
131     st.subheader("AQI Statistics")
132     st.write(data["aqi"].describe())
133
134     st.subheader("Highest AQI Record")
135     st.write(data.loc[data["aqi"].idxmax()])
136
137     st.subheader("Lowest AQI Record")
138     st.write(data.loc[data["aqi"].idxmin()])
139
140 # TRENDS
141 with tab4:
142     st.subheader("Monthly AQI Trend")
```



Explanation:

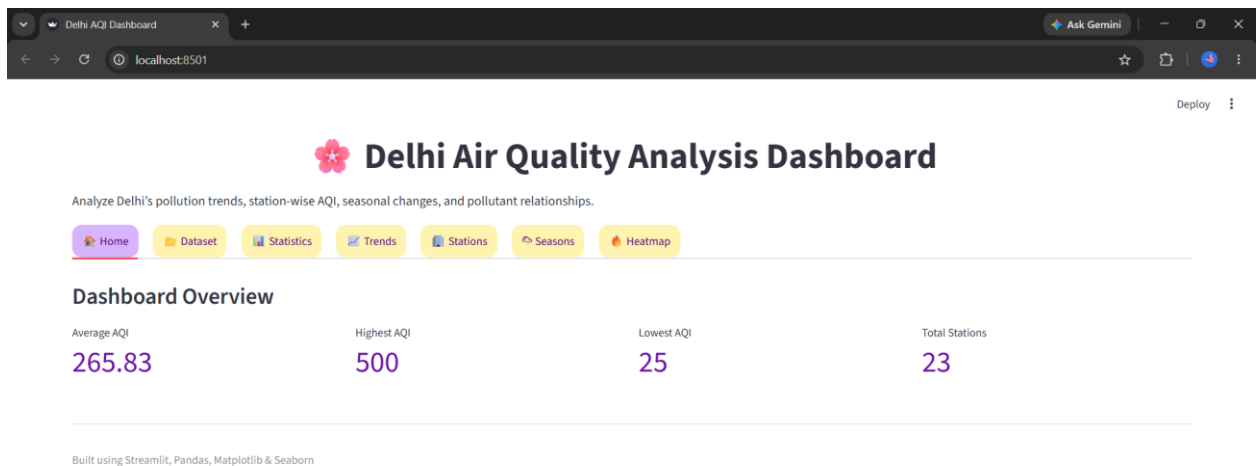
The dashboard was implemented using Python and Streamlit. Different sections such as dataset preview, AQI statistics, trend analysis, station-wise analysis, seasonal analysis, and heatmap visualization were integrated into the application.

DASHBOARD FEATURES

The developed dashboard includes the following features:

- Interactive top navigation tabs
- Dataset preview section
- AQI statistical summary
- Monthly AQI trend visualization
- Station-wise AQI analysis
- Season-wise AQI comparison
- Pollutant correlation heatmap
- Dataset download option

HOME PAGE



Explanation:

The home page provides an overview of the dashboard and displays important AQI metrics such as average AQI, highest AQI, lowest AQI, and total stations.

STATISTICS PAGE

Lowest AQI Record

	19087
datetime	2020-07-26 23:00:00
date	2020-07-26 00:00:00
year	2020
month	7
day	26
hour	23
day_of_week	Sunday
is_weekend	1
season	monsoon
city	Delhi

Built using Streamlit, Pandas, Matplotlib & Seaborn

 Delhi Air Quality Analysis Dashboard

Analyze Delhi's pollution trends, station-wise AQI, seasonal changes, and pollutant relationships.

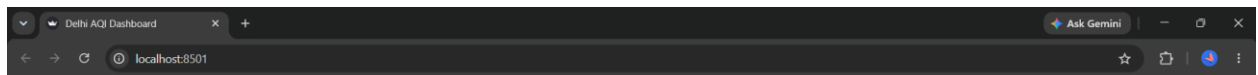
- Home
- Dataset
- Statistics
- Trends
- Stations
- Seasons
- Heatmap

AQI Statistics

	aqi	
count		201664
mean		265.8301
std		171.509
min		25
25%		103
50%		232
75%		464
max		500

Highest AQI Record

	0
--	---



Deploy

Highest AQI Record

	0
datetime	2020-01-01 06:00:00
date	2020-01-01 00:00:00
year	2020
month	1
day	1
hour	6
day_of_week	Wednesday
is_weekend	0
season	winter
city	Delhi

Lowest AQI Record

	19087
datetime	2020-07-26 23:00:00
date	2020-07-26 00:00:00
year	2020

Explanation:

This section provides statistical information such as count, mean, minimum, maximum, and quartiles of AQI values. It also highlights the highest and lowest AQI records.

TREND ANALYSIS

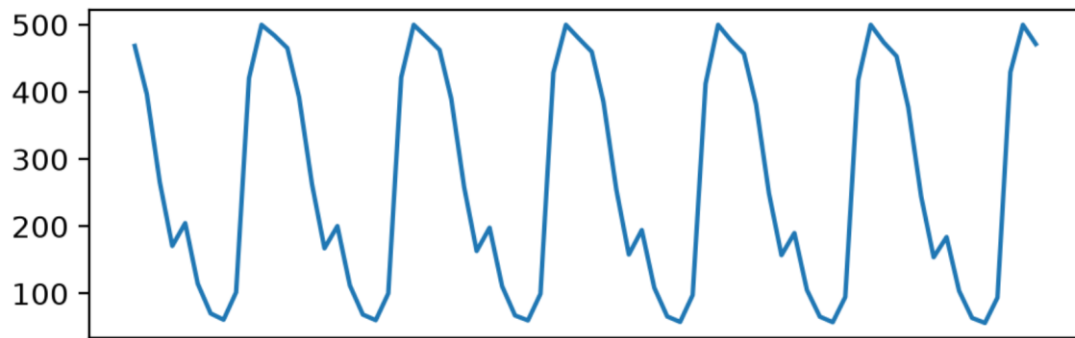


Delhi Air Quality Analysis Dashboard

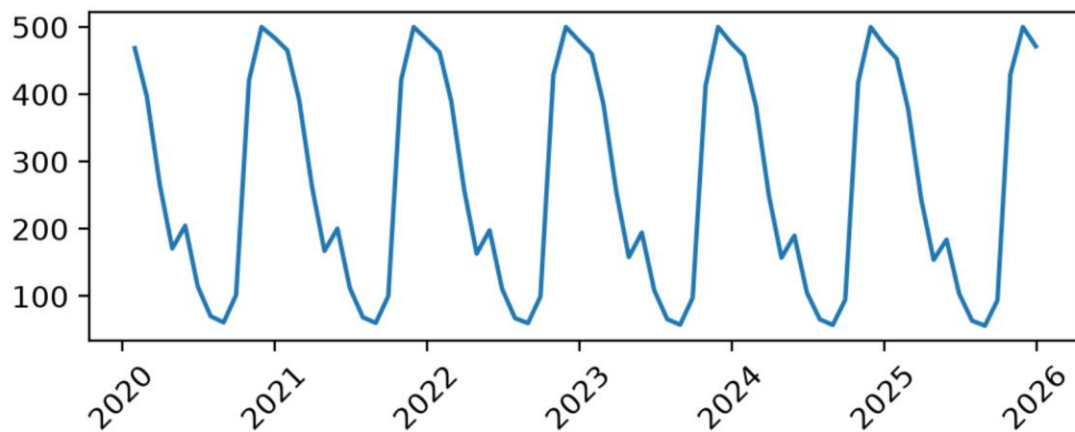
Analyze Delhi's pollution trends, station-wise AQI, seasonal changes, and pollutant relationships.

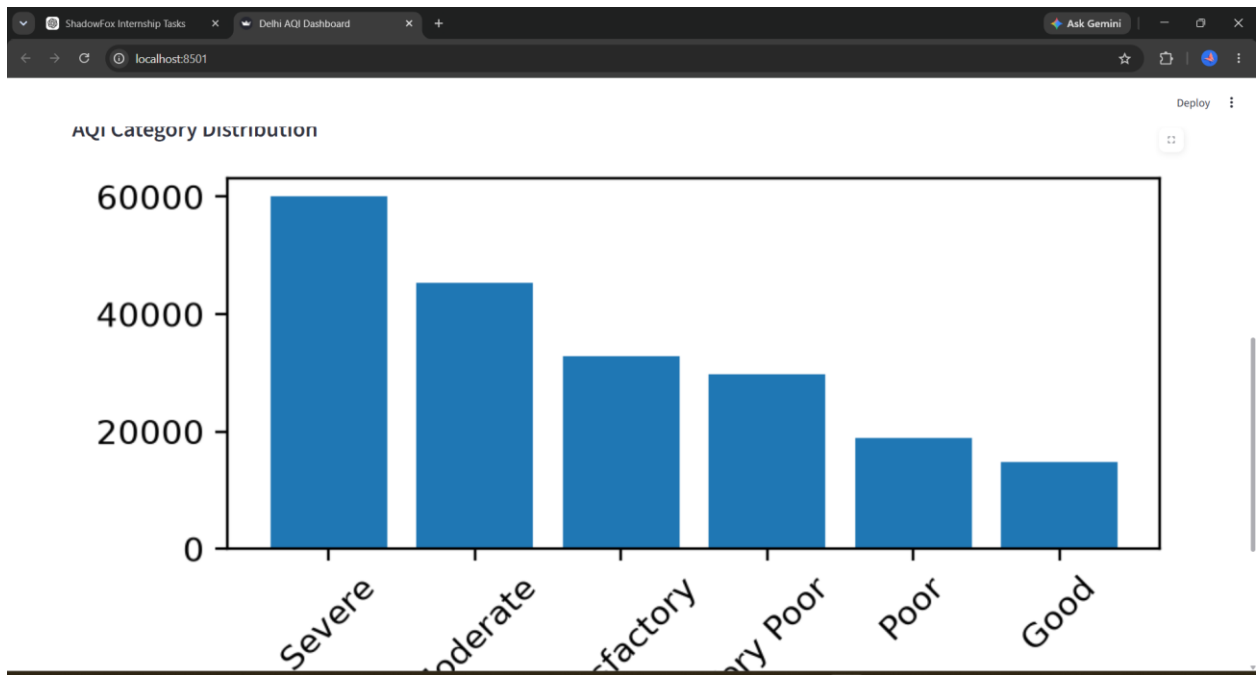
- Home
- Dataset
- Statistics
- Trends
- Stations
- Seasons
- Heatmap

Monthly AQI Trend



Monthly AQI Trend





Explanation:

This section visualizes monthly AQI trends and AQI category distribution. It helps identify pollution patterns over time.

STATION ANALYSIS

Deploy

Delhi Air Quality Analysis Dashboard

Analyze Delhi's pollution trends, station-wise AQI, seasonal changes, and pollutant relationships.

- HomeDatasetStatisticsTrendsStationsSeasonsHeatmap

Station-wise Analysis

Select Station

Anand Vihar, Delhi

Anand Vihar, DelhiJahangirpuri, DelhiWazirpur, DelhiBawana, DelhiITO, DelhiPunjabi Bagh, DelhiOkhla Phase 2, Delhi



Deploy

Delhi Air Quality Analysis Dashboard

Analyze Delhi's pollution trends, station-wise AQI, seasonal changes, and pollutant relationships.

- HomeDatasetStatisticsTrendsStationsSeasonsHeatmap

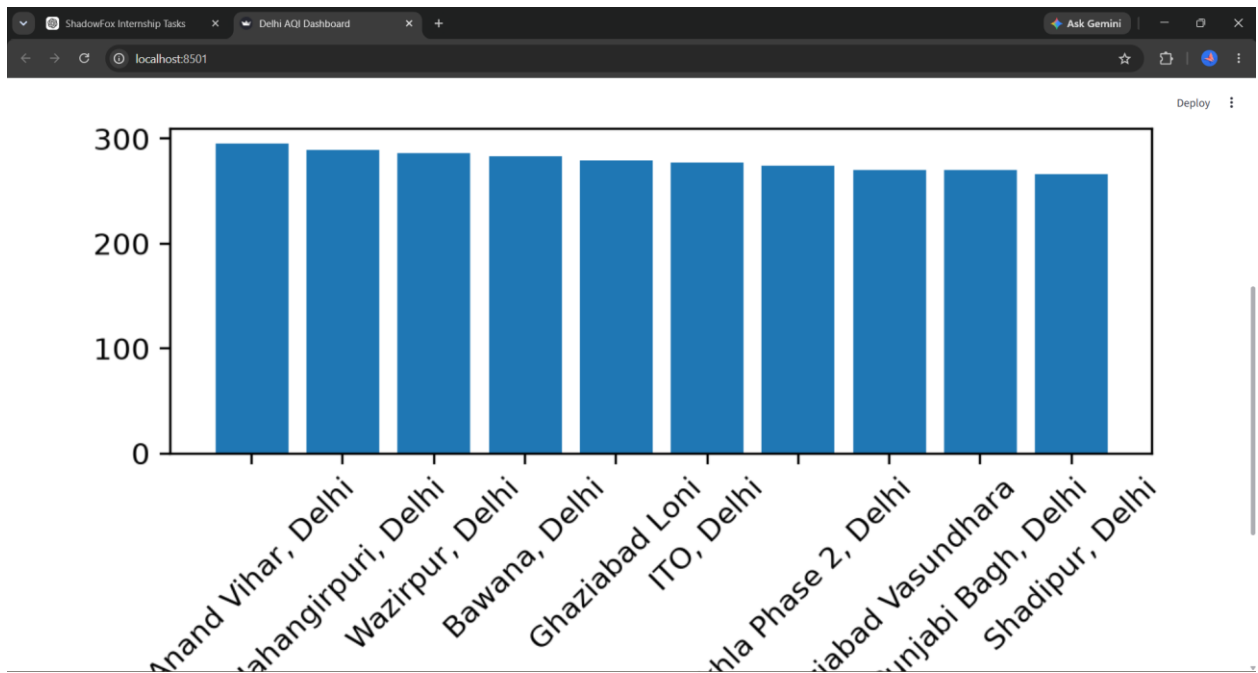
Station-wise Analysis

Select Station

Anand Vihar, Delhi

	datetime	date	year	month	day	hour	day_of_week	is_weekend	season	city	station	latitude	longitude	pm25	pm10	no2	so2	co	o3
0	2020-01-01 06:00:00	2020-01-01 00:00:00	2020	1	1	6	Wednesday	0	winter	Delhi	Anand Vihar, Delhi	28.6469	77.3164	371.8	739.4	119.6	47.7	5.19	12.3
1	2020-01-01 12:00:00	2020-01-01 00:00:00	2020	1	1	12	Wednesday	0	winter	Delhi	Anand Vihar, Delhi	28.6469	77.3164	301.1	588.8	117.9	39.3	4.32	15.8
2	2020-01-01 18:00:00	2020-01-01 00:00:00	2020	1	1	18	Wednesday	0	winter	Delhi	Anand Vihar, Delhi	28.6469	77.3164	334	602.6	150.1	36.3	7.13	14.3
3	2020-01-01 23:00:00	2020-01-01 00:00:00	2020	1	1	23	Wednesday	0	winter	Delhi	Anand Vihar, Delhi	28.6469	77.3164	403.8	841.8	142	30.3	4.9	13.2
92	2020-01-02 06:00:00	2020-01-02 00:00:00	2020	1	2	6	Thursday	0	winter	Delhi	Anand Vihar, Delhi	28.6469	77.3164	448.4	749.2	134.8	39.9	7.62	15.1

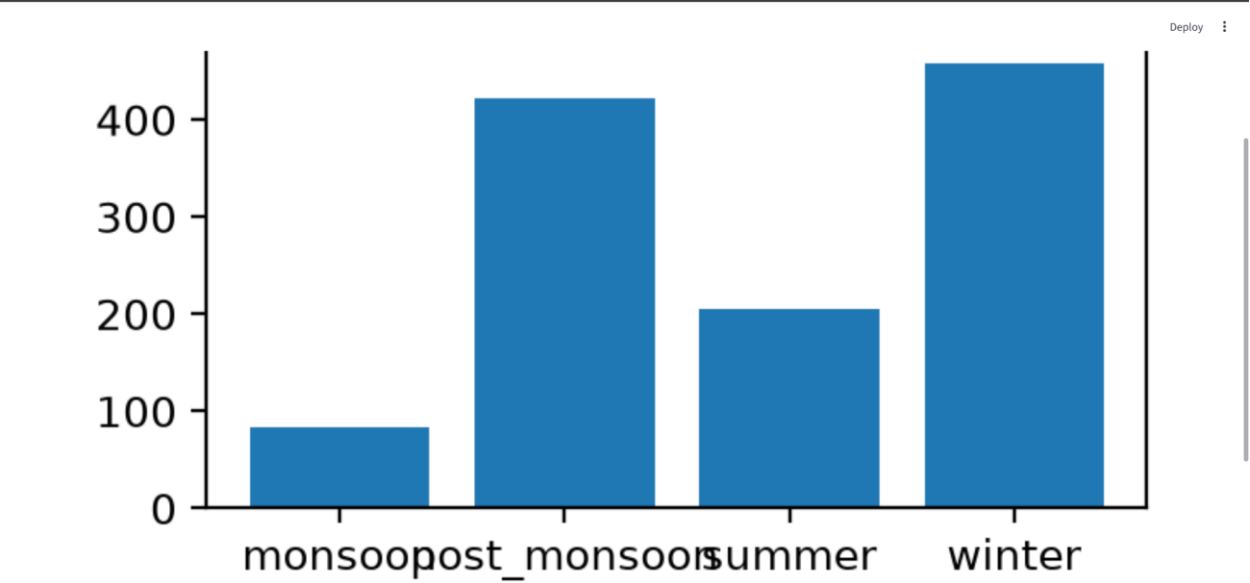
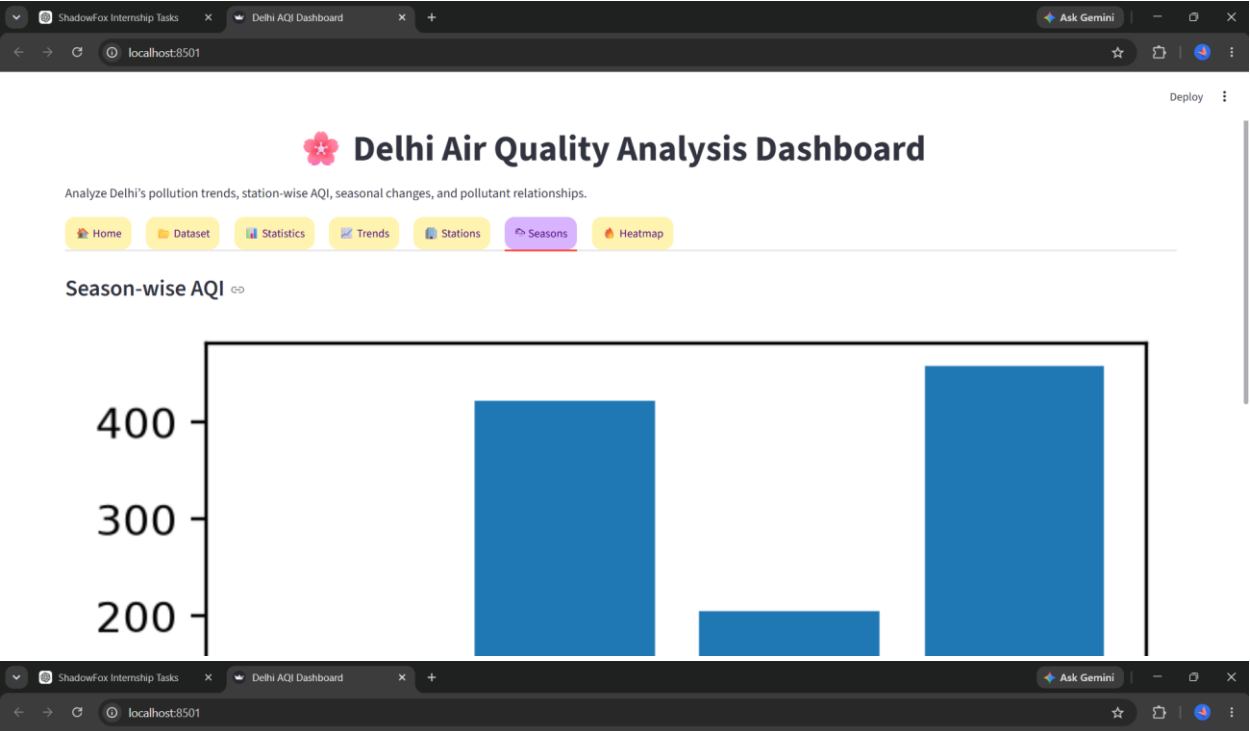




Explanation:

This section allows users to select a specific monitoring station and view AQI data for that station. It also displays the top 10 most polluted stations in Delhi.

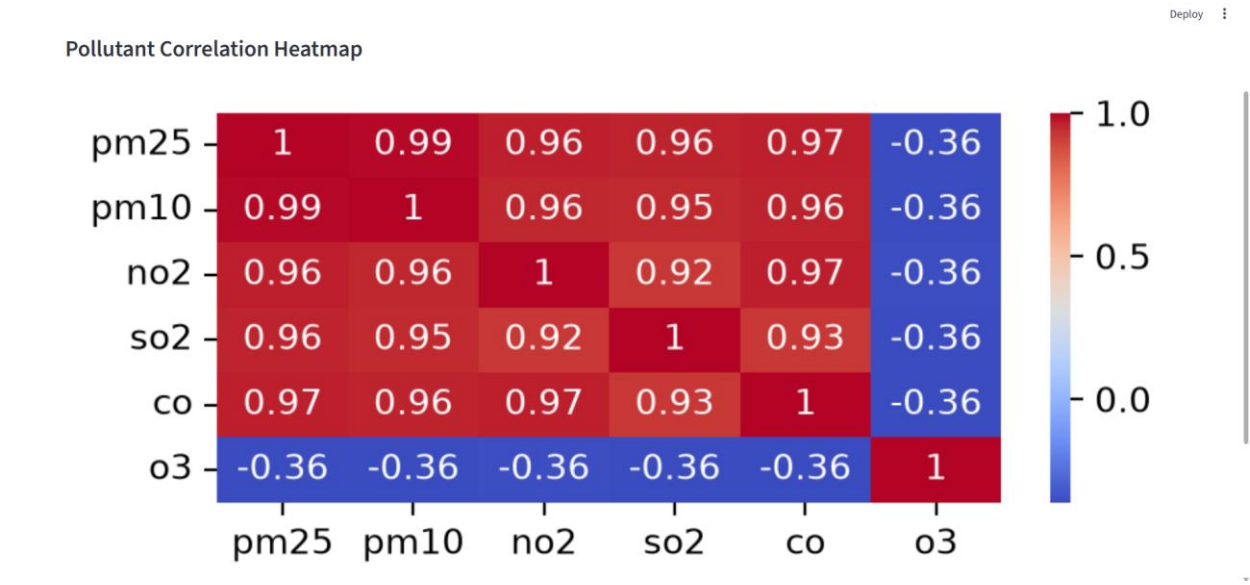
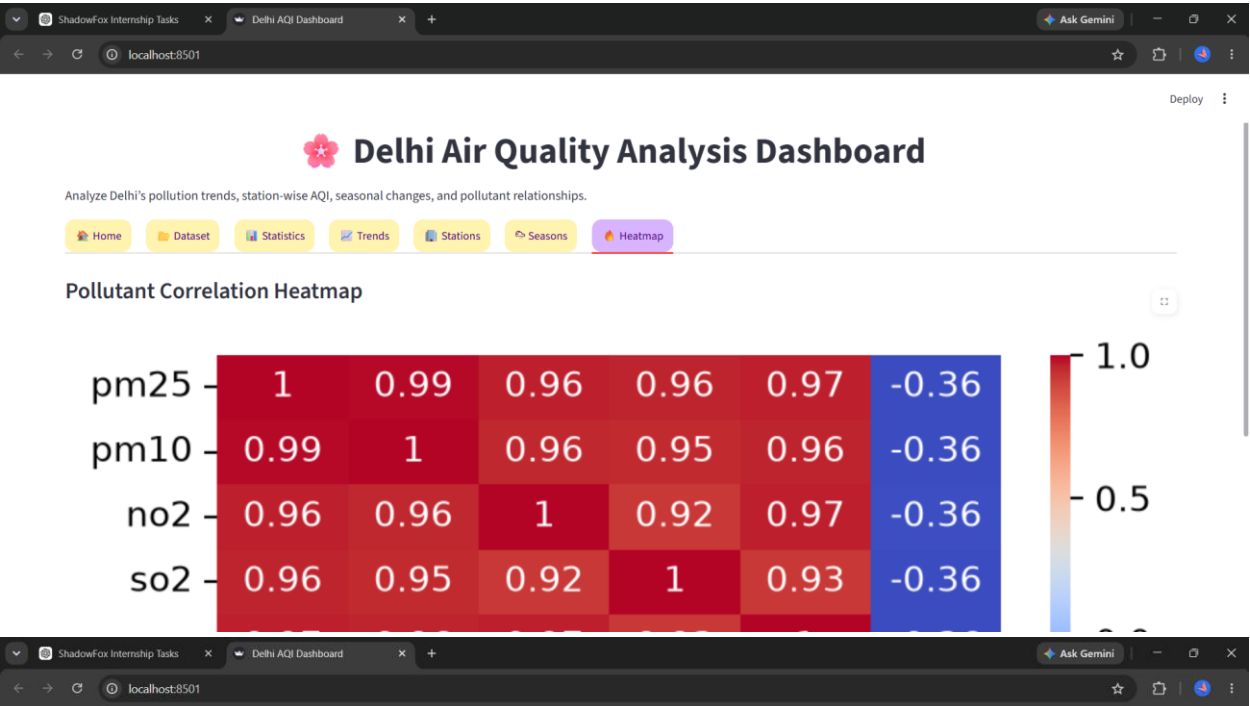
SEASON ANALYSIS



Explanation:

This section compares AQI values across different seasons. It helps understand how air pollution changes in winter, summer, monsoon, and other seasons.

HEATMAP ANALYSIS



Explanation:

The heatmap shows the correlation between different pollutants such as PM2.5, PM10, NO2, SO2, CO, and O3. It helps identify strong pollutant relationships.

CONCLUSION

This project successfully analyzed Delhi's air quality data and developed an interactive dashboard for visualization. The dashboard provides insights into AQI trends, pollution levels across stations, seasonal variations, and pollutant correlations. It helps in understanding air quality conditions and can support environmental monitoring and awareness.